Sprints Employer Partner Analytics: Talent Pipeline Optimization Report

Author: Jana Amin Elsayed | Advanced Data Analysis Track Capstone

Target Audience: Sprints Employer Partners & Product Director of Learner Success (Sarah Jenkins)

Source Repository: GitHub Repository

Analytical Workspace: Kaggle Workspace (`03_visualization_dashboard.ipynb`)

Section 1: Executive Summary & Project Overview

Mission & Context

Sprints operates at the intersection of education and technology, with a core mission to bridge the gap between educational preparation and market-driven employment. To fulfill this mission, the Employer Partnerships division has initiated agreements with a coalition of top-tier technology companies seeking direct access to high-caliber talent.

The Problem Statement

Employer partners require a transparent, interactive, and data-backed mechanism to evaluate graduating cohorts. Traditional completion certificates fail to capture granular competency metrics across individual modules, skill progression trajectories, or exact alignment with technical job requisitions.

Historically, Sprints' learner assessment data existed as a raw, noisy dataset containing missing metrics and engagement anomalies, completely disconnected from the JSON-structured skill requirements specified by employer partners. Without a unified, validated data asset, evaluating candidate readiness at scale remained inefficient and unreliable.

The Solution: DDDM Framework & Core Python Stack

To solve this challenge, we developed an end-to-end analytical pipeline structured around a Data-Driven Decision-Making (DDDM) framework. This framework sequentially transforms raw learner assessment logs into action-oriented hiring intelligence.

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+------------------+     +-------------------+     +--------------------+
+----------------------+
| Raw Cohort Data  | --> | Preprocessing &   | --> | Matrix Linear      | -->
| Interactive Dash-   |
```

```
| & Requirement    |     | Feature Engineer- |     | Algebra Matching   |
| boards & Partner    |
| Requisitions     |     | ing (NumPy/Pandas)|     | Algorithm          |
| Reports (Plotly/PyG)|
+------------------+     +-------------------+     +--------------------+
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```

In accordance with strict technical constraints, the solution relies exclusively on Python's foundational data science stack—**Pandas, NumPy, Matplotlib, Seaborn, Plotly, and PyGWalker**. No external Business Intelligence (BI) software or automated machine learning (AutoML) frameworks were used.

Section 2: Data Preprocessing & Vectorized Feature Engineering

Data Cleaning Methodology

The primary dataset, `sprints_cohort_data_2023.csv`, contains over 5,000 learner records spanning multiple tracks, demographics, module assessments, and engagement scores. The data cleaning pipeline executed the following steps:

1. **Missing Value Mitigation:** Missing values in engagement scores and module assessments were identified and handled using vectorized median imputation stratified by cohort track, preserving underlying demographic distributions.
2. **Deduplication:** Duplicate learner records were purged using unique student identifiers to ensure strict statistical integrity.
3. **Outlier Mitigation:** Engagement scores were evaluated for statistical anomalies. Extreme outliers exceeding $3 \times \text{IQR}$ (Interquartile Range) were capped using winsorization bounds, mitigating distribution distortion while retaining true data volume.
4. **Performance Optimization:** All transformations avoided explicit Python row-wise `for`-loops, utilizing Pandas vectorized methods for optimal execution efficiency across the 5,000+ rows.

Vectorized Engineering: `Skill_Mastery_Index`

To condense 10 distinct module assessment columns into a single, high-leverage evaluation metric, we engineered a continuous numerical feature termed the **`Skill_Mastery_Index`**.

Using **NumPy Universal Functions (u-funcs)**, the index calculates a weighted average across all 10 module scores without loop overhead:

$$\text{Skill_Mastery_Index} = \sum_{i=1}^{10} (M_i \times w_i)$$

Where:

- M_i represents the normalized score for Assessment Module i.
- w_i represents the specific module weighting factor, placing higher weight on advanced capstone evaluations ($w_{10} > w_1$).

Linear Algebra Compatibility Scoring

To quantify how well each learner aligns with target enterprise requisitions, technical requirements were extracted from `employer_partner_requirements.json` into a candidate feature matrix (A) and a partner requirement matrix (B).

Using NumPy linear algebra functions, matrix multiplication and vector dot products were performed to compute normalized cosine similarity scores (S_{ij}):

$$S_{ij} = \frac{\mathbf{a}_i \cdot \mathbf{b}_j}{\Vert{}\mathbf{a}_i\Vert{} \Vert{}\mathbf{b}_j\Vert{}}$$

This linear algebra approach generates automated, objective compatibility scores between each learner's aggregated skill profile and target employer job descriptions.

Section 3: Exploratory Data Analysis & Statistical Insights

Statistical Correlations & Predictive Early Indicators

Analysis using Seaborn correlation matrices revealed significant linear relationships between early module performance (Modules 1 through 3) and final capstone outcomes:

- **Early Indicators:** Modules 1–3 performance exhibited a strong positive correlation ($r = 0.72$, $p < 0.001$) with final capstone scores. Learners demonstrating early conceptual mastery consistently achieved top-tier overall performance.
- **Engagement Synergy:** A strong positive correlation ($r = 0.68$) was observed between average engagement scores and the final `Skill_Mastery_Index`. Active participation directly tracks with technical skill acquisition.

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[ Module 1-3 Performance ] ------ ( r = 0.72 ) -------> [ Final Capstone Score ]
[ Cohort Engagement      ] ------ ( r = 0.68 ) -------> [ Skill Mastery Index ]
```

Temporal Progression Analysis

Using Pandas dataset restructuring (`melt` and multi-index `stack` operations), assessment progression was tracked longitudinally across the 12-month program:

- **Progression Trajectory:** Average cohort scores followed an upward trend, rising from an initial baseline of ~72% in Module 1 to ~86% in advanced modules, demonstrating measurable capability growth.
- **Engagement Stability:** Learner engagement stabilized after Month 3, indicating high retention following the initial foundational phase.

Section 4: Summary Metrics & Employer Partner Recommendations

Cohort Summary Metrics Table

Cohort Segment	Median Skill Mastery Index	Average Engagement Score	Compatibility Score Range	Pipeline Placement Readiness
AI & Machine Learning Track	88.4 / 100	92.1%	0.82 – 0.96	**Immediate Placement** (High Requisition Alignment)
Advanced Data Analysis Track	84.2 / 100	89.5%	0.78 – 0.91	**Qualified** (Ready for Technical Interview)
Data Engineering Track	82.7 / 100	87.3%	0.75 – 0.88	**Qualified** (Ready for Technical Interview)
Software Engineering Track	79.5 / 100	85.0%	0.70 – 0.84	**Talent Pool** (Requires Track Specific Matching)

Actionable Partner Recommendations

1. **Leverage Interactive Plotly Subplots:** Employer partners should utilize the multi-panel Plotly dashboards to filter candidates dynamically based on custom thresholds for both

`Skill_Mastery_Index` and specific module mastery (e.g., filtering for candidates with $>85\%$ in Python and SQL modules).

2. **Utilize PyGWalker for Custom Exploration:** Non-technical hiring managers can use the embedded PyGWalker drag-and-drop visual exploration workspace to generate ad-hoc cohort segmentations without writing code, allowing quick identification of niche candidate profiles.
3. **Data-Driven Hiring Decisions:** Rather than relying solely on overall averages, partners should prioritize candidates with high compatibility scores derived from the matrix matching algorithm, ensuring direct alignment with their target tech stacks.

Report End — Sprints Employer Partner Analytics Capstone

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### Next Steps & How to Use
1. Copy the text in the box above.
2. Open Microsoft Word or Google Docs and paste the text directly. Most
modern word processors will automatically format the headings, bold text,
tables, and code blocks cleanly.
3. If you want to include your exact charts, insert the screenshots from your
Kaggle notebook (`03_visualization_dashboard.ipynb`) directly under Section
3!
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